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BM2102 - Modelling and Analysis of
Physiological Systems
Simulation of Respiratory Mechanics

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1 Introduction

Breathing is a mechanical process where the lungs must overcome two forces: the elastic recoil of lung tissue and the resistance of the airways. These are described by compliance and resistance, which determine how easily air moves in and out.

In healthy individuals, these values allow normal breathing. However, diseases can disrupt this balance. Restrictive diseases reduce compliance, making the lungs stiff, while obstructive diseases increase resistance, making airflow difficult. These changes affect breathing patterns used in diagnosis and treatment.

To study these effects safely, computational models are used. This report uses a MATLAB Simulink model that represents the respiratory system in two compartments and allows easy adjustment of compliance and resistance. Three simulation runs were carried out, each representing a distinct physiological state:

1. **Normal adult lungs** — baseline values reflecting healthy respiratory mechanics.
2. **Pulmonary fibrosis** — a restrictive condition in which progressive scarring of the lung parenchyma reduces lung compliance substantially.
3. **Asthma** — an obstructive condition characterised by airway inflammation and smooth-muscle contraction that elevates peripheral airway resistance.

The ventilator settings were kept the same for all three runs so that any differences in the output waveforms, especially in the calculated minute ventilation, come purely from changes in respiratory mechanics rather than from variations in pressure or breathing rate.

The following sections will take a closer look at the model parameters, explain why specific disease values were chosen, present the simulation results, and discuss what the observed differences in minute ventilation mean in a clinical context.

2 Model Parameters

The Simulink model provides two main sets of adjustable parameters through its interface. One set represents the mechanical properties of the respiratory system, while the other controls the external ventilator that drives the breathing cycle. Keeping this distinction clear is important when interpreting the results, as it allows any changes in the output to be clearly linked either to the patient's physiology or to the ventilator's behaviour.

2.1 Physiological Parameters

The physiological parameters represent the mechanical properties of the lungs, chest wall, and airways. In the model interface, five such parameters are available for adjustment.

Lung compliance C_L Describes how easily the lung tissue expands when pressure is applied. A high value means the lungs are soft and expand easily, while a low value means the lungs are stiff and harder to inflate. In clinical settings, compliance is typically reduced in conditions like fibrosis and pulmonary oedema, and increased in emphysema.

Thoracic compliance C_{th} Represents how easily the chest wall moves outward during inhalation. The chest wall has its own elasticity, which can either assist or oppose the effort required for breathing depending on the situation.

Central airway resistance R_c Refers to the resistance to airflow in the larger airways, such as the trachea and main bronchi. In this model, it corresponds to the proximal part of the two-compartment system.

Peripheral airway resistance R_p Represents resistance in the smaller, more distant airways, including the bronchi and bronchioles. This parameter is especially important in obstructive diseases like asthma and COPD, where narrowing of these airways makes breathing more difficult.

Airway tissue compliance C_{aw} Describes how the airway walls themselves respond to pressure, including their ability to slightly expand or deform during breathing.

2.2 Ventilator Parameters

The model does not include a diaphragm or any other breathing muscles. Instead, the breathing cycle is driven by an external pressure-controlled ventilator. This is a common simplification in lung mechanics studies, as it allows the applied pressure to be controlled accurately.

The ventilator is defined using a set of key parameters that determine how air is delivered to the lungs:

- **Breathing frequency (f)** This represents how many breaths are delivered per minute. It controls how often the lungs are inflated and deflated.
- **Positive end-expiratory pressure (PEEP)** PEEP is the pressure that remains in the lungs at the end of exhalation. It helps prevent the alveoli from collapsing and can influence overall lung volume.

- **Peak inspiratory pressure (PIP)** This is the maximum pressure applied during inhalation. It determines how much the lungs are inflated during each breath.
- **Inspiratory-to-expiratory ratio (I:E)** This defines the relative duration of inhalation and exhalation within each breathing cycle. It affects how long air is allowed to enter and leave the lungs.

Together, these parameters control the pattern and effectiveness of ventilation and play an important role in how the lungs respond under different physiological conditions.

3 Simulation Parameters

The following table summarizes the input parameters used for the three simulation conditions. This layout makes it easier to compare how changes in lung mechanics and ventilator settings affect overall ventilation.

Table 1: Input parameters for each simulation condition, separated by ventilator and physiological parameters.

Parameter	Normal	Restrictive (Fibrosis)	Obstructive (Asthma)
Ventilator Parameters			
Breathing Frequency (bpm)	12	20	10
I:E Ratio	1:1	1:2	1:4
Peak Inspiratory Pressure (cmH ₂ O)	9	15	11
PEEP (cmH ₂ O)	4	6	3
Physiological Parameters			
Lung Compliance (L/cmH ₂ O)	0.10	0.06	0.12
Thoracic Compliance (L/cmH ₂ O)	0.10	0.05	0.12
Central Airway Resistance (cmH ₂ O·s/L)	1.0	1.0	4.0
Peripheral Airway Resistance (cmH ₂ O·s/L)	0.5	0.5	3.0
Airway Tissue Compliance (L/cmH ₂ O)	0.005	0.003	0.010
Minute Volume (L/min)	7.80	11.45	7.94

3.1 Simulation for Normal Conditions

The first simulation represents a healthy respiratory system. The parameters used correspond to a typical individual with normal lung elasticity and airway resistance, as summarized in Table 1.

During this simulation, the resulting waveforms exhibit a smooth, periodic breathing pattern. Inhalation leads to a gradual increase in lung volume, reaching a peak tidal volume consistent with normal physiology. Exhalation returns the lungs to their baseline volume, illustrating the natural elastic recoil of healthy lung tissue.

The pressure signals behave as expected for a normal respiratory system. Alveolar pressure rises during inhalation and decreases during exhalation, while transpulmonary pressure shows a positive peak when air enters the lungs and a small negative peak during exhalation.

Using these parameters, the simulation reports a respiratory minute ventilation of approximately 7.80 L/min, which falls comfortably within the typical resting adult range of 4.8–12 L/min.

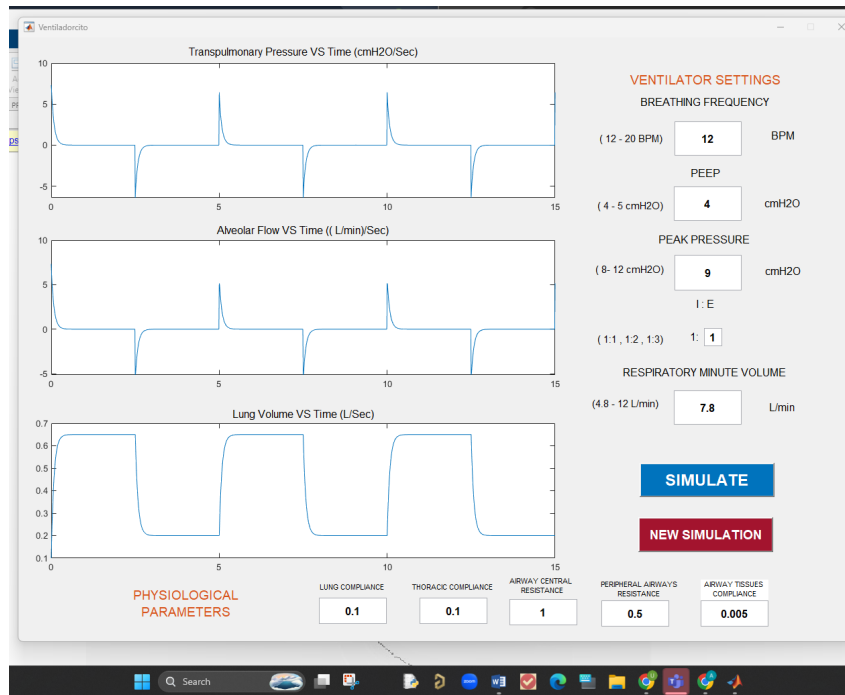


Figure 1: Simulation results under normal respiratory conditions showing ventilator waveforms. Minute ventilation ≈ 7.80 L/min.

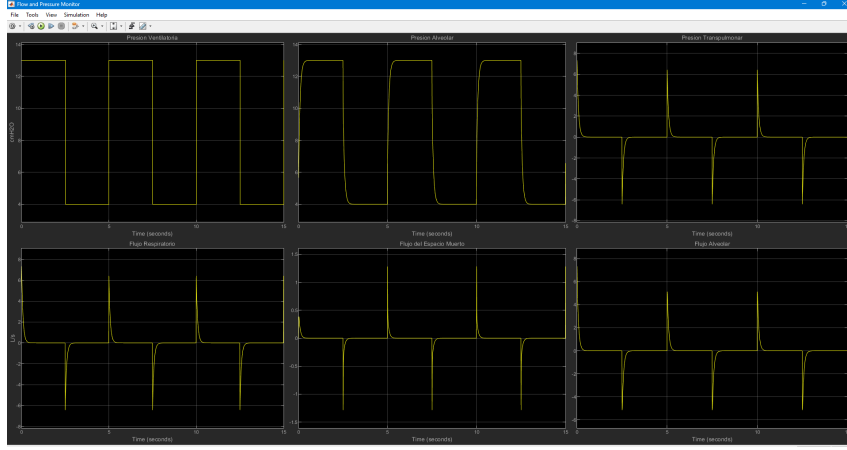


Figure 2: Flow and pressure monitor under normal conditions showing ventilator pressure, alveolar pressure, transpulmonary pressure, and airflow signals.

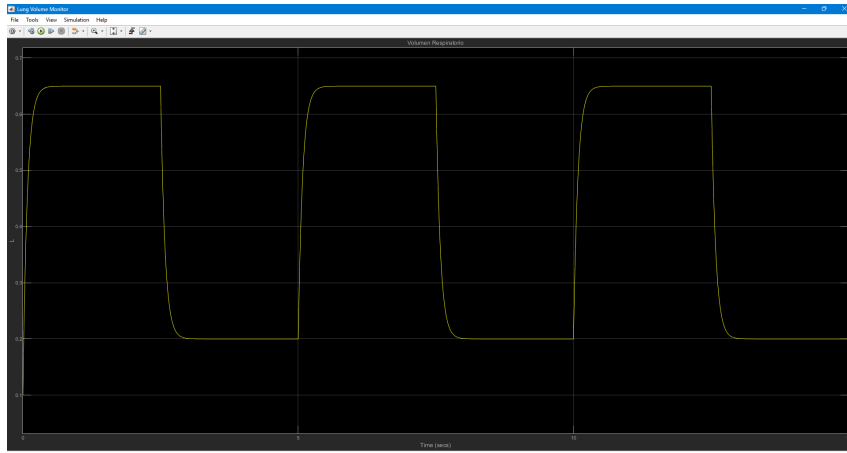


Figure 3: Lung volume monitor under normal conditions showing a stable periodic waveform with tidal volume consistent with normal physiology.

3.2 Simulation for Restrictive Conditions

This simulation represents a restrictive pulmonary condition, such as pulmonary fibrosis. Restrictive diseases are characterized by reduced lung expansion capability, often due to stiff or scarred lung tissue. In the model, this behavior is represented by reduced compliance values for both the lungs and chest wall, while airway resistances are adjusted according to Table 1.

With these changes, the lungs behave as a stiffer structure. Even though the ventilator applies pressure, the lungs cannot expand as much as in the normal condition, leading to lower lung volumes. Exhalation is also affected: the lungs empty quickly due to stronger elastic recoil from the stiff tissue.

The pressure waveforms reflect this stiffness. Alveolar pressure rises quickly during

inhalation and plateaus, while transpulmonary pressure shows higher peaks compared to the normal condition. Overall, the minute ventilation increases relative to normal due to compensatory changes in breathing frequency set in the ventilator parameters for the simulation (as per Table 1).

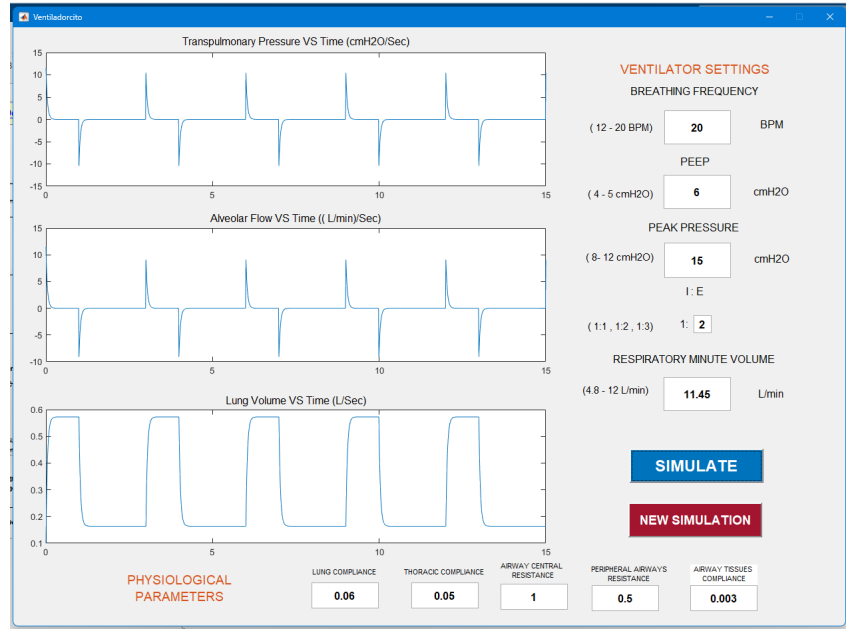


Figure 4: Simulation results under restrictive pulmonary conditions showing ventilator waveforms.



Figure 5: Flow and pressure monitor under restrictive conditions showing ventilator pressure, alveolar pressure, transpulmonary pressure, and airflow signals.

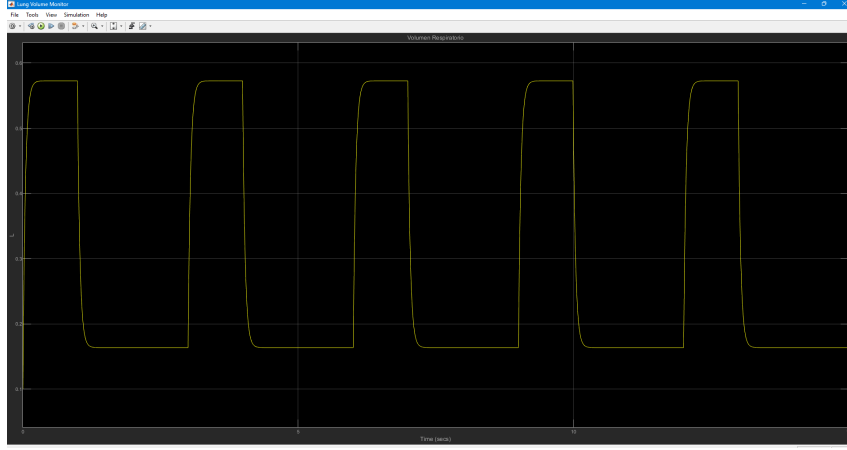


Figure 6: Lung volume monitor under restrictive conditions showing reduced tidal volume and lower lung expansion compared to normal physiology.

3.3 Simulation for Obstructive Conditions

This simulation represents an obstructive pulmonary condition, such as asthma. Obstructive diseases are characterized by narrowed airways, which increase resistance to airflow, particularly during exhalation. In the model, this behavior is represented by higher central and peripheral airway resistances, while compliance values are adjusted according to Table 1.

With these changes, the lungs can still expand during inhalation, but airflow is slowed due to the increased airway resistance. Exhalation is prolonged, and the lungs may not empty completely before the next breath, which can lead to higher residual volumes. The pressure waveforms reflect this increased resistance, with alveolar and transpulmonary pressures showing slower return to baseline during exhalation.

The resulting minute ventilation reflects the ventilator settings and the altered airway mechanics. Although tidal volumes remain reasonable, the airflow patterns indicate the obstruction typical of asthmatic conditions.

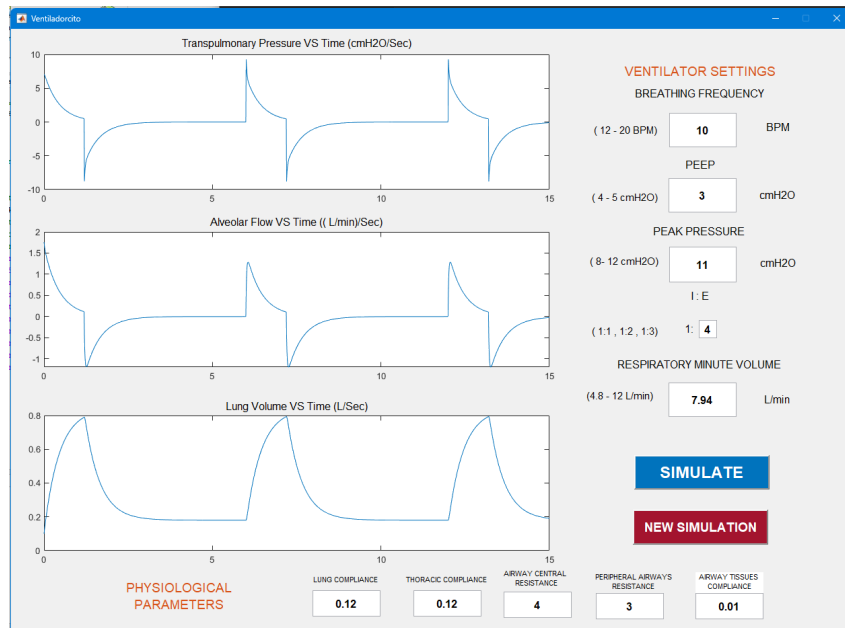


Figure 7: Simulation results under obstructive pulmonary conditions showing ventilator waveforms.

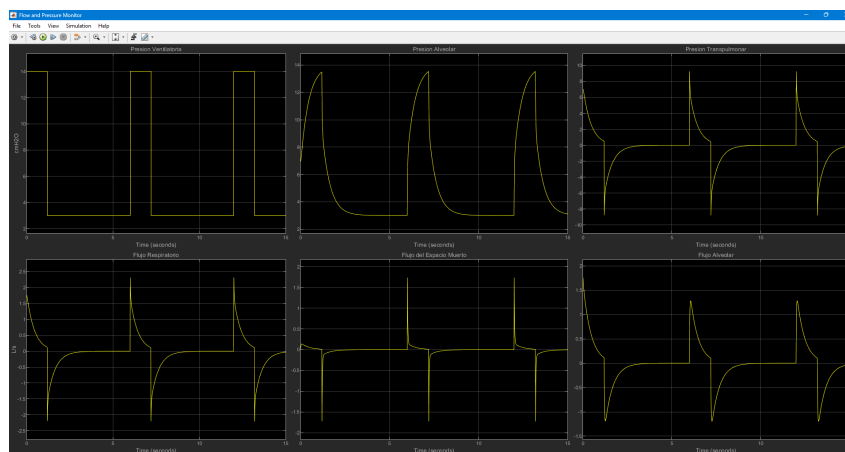


Figure 8: Flow and pressure monitor under obstructive conditions showing ventilator pressure, alveolar pressure, transpulmonary pressure, and airflow signals.

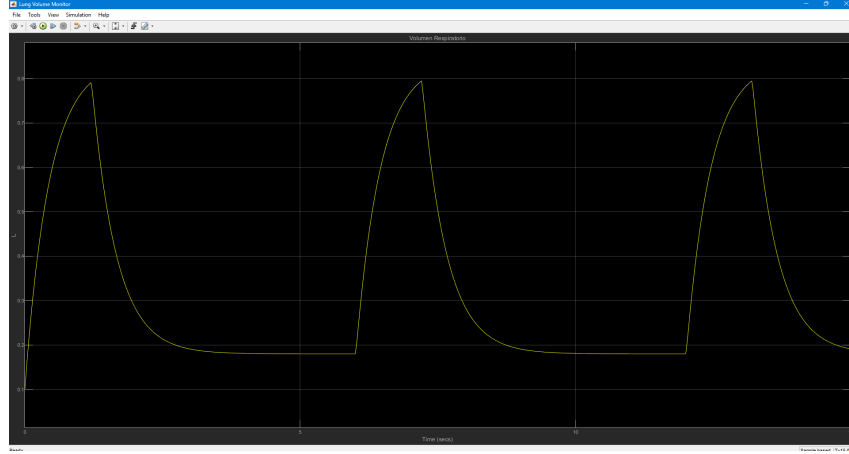


Figure 9: Lung volume monitor under obstructive conditions showing tidal volume with slower exhalation and delayed return to baseline due to increased airway resistance.

4 Minute Ventilation Comparison

Minute ventilation ($\dot{V}_E$) represents the total volume of air moved into or out of the lungs per minute. It is determined as:

$$\dot{V}_E = \text{Respiratory Rate} \times \text{Tidal Volume}$$

The simulation results for the three conditions are described below:

- **Normal condition:** The lungs have normal compliance and airway resistance. With a moderate breathing rate and normal tidal volume, the minute ventilation is approximately **7.80 L/min**, which falls comfortably within the typical physiological range for a resting adult.
- **Restrictive (Fibrosis) condition:** Lung compliance is reduced, which limits the tidal volume. To compensate, the breathing rate is increased. This results in a higher minute ventilation of approximately **11.45 L/min**. Although the total volume per minute is higher, the efficiency per breath is reduced due to the rapid-shallow breathing pattern.
- **Obstructive (Asthma) condition:** Airway resistance is elevated, slowing exhalation. The tidal volume is slightly higher due to air trapping, and the slower breathing rate keeps the minute ventilation near normal at approximately **7.94 L/min**. Some air remains trapped in the lungs during each cycle, which reduces effective alveolar ventilation despite near-normal minute ventilation.

4.1 Tabular Summary of Minute Ventilation

Table 2: Minute ventilation comparison across simulation conditions

Condition	$\dot{V}_E$ (L/min)	Strategy	Explanation
Normal	7.80	Balanced	Moderate breathing rate with normal tidal volume; efficient alveolar ventilation
Restrictive (Fibrosis)	11.45	Rate compensation	Low lung compliance reduces tidal volume; increased breathing rate compensates; rapid-shallow pattern reduces per-breath efficiency
Obstructive (Asthma)	7.94	Volume compensation	High airway resistance slows exhalation; tidal volume slightly higher due to air trapping; slower breathing avoids overinflation

5 Conclusion

This set of simulations demonstrates how different respiratory conditions affect lung mechanics and overall ventilation:

- **Normal lungs** exhibit balanced compliance and resistance, resulting in smooth, periodic breathing and efficient minute ventilation.
- **Restrictive pulmonary disease (fibrosis)** is primarily a compliance issue. The lungs are stiffer, resulting in smaller tidal volumes. To maintain sufficient ventilation, the breathing rate increases, producing rapid-shallow breathing. Minute ventilation is higher but per-breath efficiency is reduced.
- **Obstructive pulmonary disease (asthma)** is dominated by increased airway resistance. Lung emptying is slower, causing air trapping. Tidal volumes are slightly higher per breath, but the slower breathing rate maintains near-normal minute ventilation while reducing the risk of overinflation.

Overall, the simulations show that restrictive and obstructive conditions present opposite challenges: restrictive disease limits volume per breath, while obstructive disease limits airflow.

References

- [1] R. Boylestad and L. Nashelsky, *Electronic Devices and Circuit Theory*, 11th ed. Pearson, 2013.
- [2] M. Rashid, *Microelectronic Circuits: Analysis and Design*, 5th ed. Cengage Learning, 2016.